SAI TEJA ADUSUMILLI

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Summary: Data Scientist skilled in Python, SQL, ML, building predictive models and analytics pipelines to generate insights and support business decisions.

EDUCATION

California State University, Sacramento — Sacramento, CA
Master of Science in Business Analytics (GPA 3.9/4.0) — Aug 2024 – Dec 2025

Krishna University — Vijayawada, India
Master of Science in Physics (GPA 4.0/4.0) — July 2017 – April 2019
Bachelor of Science in Mathematics, Physics, and Computer Science (GPA 3.99/4.0) — July 2014 – April 2017

SKILLS

- **Programming & Data Analysis:** Python(scikit-learn, PySpark, Pandas, NumPy), R, SQL, HTML
- **Statistical/ML Techniques:** Regression, Decision Trees, Clustering, Ensemble Methods, Neural Networks, NLP
- **Data Visualization & Analytics:** Tableau, Power BI, Matplotlib, Seaborn, Plotly, Streamlit
- **Cloud & Development Tools:** AWS (S3, EC2, SageMaker), LLM APIs, Git, Cursor, VS Code
- **Data Engineering & Databases:** MySQL, PostgreSQL, Firebase, ETL/ELT Pipelines
- **Business Skills:** Financial Accounting, Risk Management, Banking Regulations, Customer Relationship Management (CRM)

EXPERIENCE

California State University, Sacramento — Sacramento, CA
Volunteer Research Assistant - Generative AI Project — Aug 2024 – Jan 2025

- Applied BERTopic to cluster 2,000+ research articles, reducing them to 1,300 high-relevance papers for targeted analysis.
- Developed a Python pipeline using TF-IDF and rule-based NLP to classify articles under STS framework for semantic categorization.

State Bank of India — Bangalore, India
Associate — June 2022 – Jan 2023

- Analyzed customer financial data and loan performance trends across 60+ applications worth $1M to support credit approval decisions and reduce default risk.
- Led a Risk-Focused Internal Audit (RFIA) across loan and branch operations, analyzing 8 years of records to assess regulatory adherence, identify control gaps, and mitigate compliance risks; achieved an 'A' rating with zero audit findings.

Junior Associate — Jan 2021 – May 2022

- Analyzed customer transaction patterns to identify cross-selling opportunities, supporting targeted marketing strategies that increased investment and lending product sales by 35%.

PROJECTS

Credit Risk Insights Dashboard | *Python (LightGBM, XGBoost, CatBoost, Random Forest), Streamlit*

- Developed a credit-risk prediction model using 27M+ loan records, achieving 0.78 AUC-ROC and 69% recall to identify high-risk applicants and support data-driven lending decisions.
- Deployed the model in a real-time Streamlit dashboard with automated probability-of-default scoring, SHAP-based explainability, and threshold simulations to support credit approval strategy.

Database & Data Warehouse Development for a Property Management Company | *SQL, ERDPlus, Tableau*

- Developed a MySQL database and data warehouse with ETL pipelines using advanced SQL queries and stored procedures to manage leasing and tenant data and support business reporting.
- Built business intelligence reports using Tableau, delivering actionable insights to optimize property management strategies.

SBA Loan Default Prediction & Profit Optimization | *Python (Decision Trees, Ensemble Methods, Regression, Neural Networks)*

- Built machine learning models on 890K SBA loan records to analyze risk exposure using a cost-sensitive profit framework (+5% repaid, -25% default) to predict defaults and improve profitability.
- Identified Bagging as the top-performing model, estimating $2.64B in net profit with a 0.4 default probability cutoff and approval of 76% lowest risk applicants to maximize lender returns.

Fake News Detection using NLP | *NLP (n-gram, Sentiment and Stylometric Analyses, Topic Modeling (LDA))*

- Built an NLP pipeline to detect misinformation patterns in news transcripts, identifying statistically significant linguistic differences ($p<0.05$) where fake content showed 26% lower vocabulary complexity and stronger fear-based language.

Research Paper Summarizer | *Python, LLMs, OpenAI API*

- Built an LLM-powered research paper summarizer leveraging prompt engineering and GPT-5 nano to automate literature survey generation, extracting structured metadata from multiple PDFs via JSON outputs and generating relevance-scored Excel reports.

CERTIFICATIONS

- AWS Certified Cloud Practitioner — Amazon Web Services | Feb 2026